PRODUCT CONCEPT PACK

Smart Mailbox

> **Our concept in one sentence:** A low-maintenance, weather-resistant sensor inside the roadside mailbox, paired with a simple mobile companion app, tells residents in real time whether mail has arrived, so they only walk out when the trip is worth making.

1. Our Product Concept

The problem (from our Opportunity Brief)	Our answer
Roadside mailboxes sit far from the door. People walk out daily just to check, and most trips end at an empty box. In ice, darkness, rain or storms the walk becomes a risk.	**The mailbox itself reports when a trip is worthwhile.** Residents stop choosing between a risky, unnecessary walk and missing important mail.

Decision	Our proposed concept
Primary user	Older adults (65+), especially living alone, and people with reduced mobility in rural or detached homes. Secondary: busy households with long driveways.
Value proposition	Know from indoors whether mail has arrived: no empty trips, no risky “just in case” walks, no letters left unseen for days.
Product format	Battery-powered module inside the existing mailbox with a built-in mobile connection (no home Wi-Fi needed) + simple mobile companion app with push alerts (optional SMS backup).
Initial focus	Version 1 does one job: a reliable “Mail arrived” / “Empty” status and an instant alert.

Essential capabilities

What it must do	Why this matters
1. Detect whether mail is inside the box	The module notices when mail is resting inside the mailbox, not just when the lid moves. Turns guessing into certainty: the heart of the value.
2. Send an instant phone alert	Users know from anywhere. Built-in mobile connection (no home Wi-Fi needed).
3. Show a clear status in the app	Large “Mail arrived” / “Empty” display and battery level, readable at a glance by non-technical users.
4. Reset automatically when emptied	Status stays on “Mail arrived” as long as mail is inside, and switches back to “Empty” as soon as it is taken out. No buttons to press, nothing to clear manually.
5. Work outdoors all year	Weather-resistant, long battery life through winter, rain and heat. Rural users cannot easily service it.

Initial boundaries

In scope (Version 1)	Out of scope (for now)
• Detecting whether mail is inside the box • Instant alert (app push, SMS backup) • Simple “Mail arrived” / “Empty” status • Automatic reset when emptied, battery alerts	• Camera or video surveillance • Automatic door-opening or mail retrieval • Identifying mail type or content • Payment or parcel-collection integration • Weather or road-safety advice • Multi-mailbox dashboards • Any extra work for Posti delivery staff

> **Why so narrow?** A focused alert keeps cost, complexity and risk low, so we can validate the core concept before adding anything else.

2. Users and Proposed Experience

Mailbox behaviour today

1×+ / day how often target users check the mailbox (assumption)	**~1× / month** consumer letters and cards reaching a home (Posti)	**–12%** addressed letter volumes in Finland, 2023 (Posti)	**50–100 m** typical distance from door to rural mailbox

What still arrives is mostly bills, official letters, magazines and advertising, at irregular times. **Result: many trips per week to an empty box.**

Who struggles most	Why the routine is hard
Older people (65+), living alone	Check out of habit; nobody to share the walk with.
People with limited mobility or weak balance	Each trip takes real effort and planning.
Rural residents with long driveways	One empty check costs several minutes and a lot of effort.
Busy residents (secondary)	*A small waste of time rather than a barrier.*

When it is icy, dark, stormy, rainy or very hot, people…	Consequence
1. Delay: check only every few days	Bills, deadlines or official letters are missed
2. Time shift: check only in daylight or the warmest hour	Often before the post has even arrived
3. Rely on others: ask family, a neighbour or a carer	Depends on someone being nearby and willing
4. Take the risk: walk out anyway on suspicion	Danger accepted just to end uncertainty

These routines are what our product aims to replace. Prototype testing will evaluate how effectively instant arrival alerts replace empty trips, postponed checks and risky “just in case” walks.

Primary persona



Aino Virtanen

72 · Retired nurse

Rural Finland · Lives alone

"I want to know if I have mail, without walking out into the cold to check."

HOME & SURROUNDINGS

- Detached house; mailbox ~80 m away at the end of a sloping gravel driveway
- Son lives two hours away

DAILY MAILBOX HABIT

- Walks out once a day after lunch, whether or not the post has come
- Box is usually empty or only advertising

WHEN THE WEATHER TURNS

- Ice or darkness: delays the check by a day or two
- Asks her son to phone a neighbour to check
- Sometimes goes anyway, fearing missed bills or letters from the municipality or healthcare

CONSTRAINTS

- Weaker balance, uses a walking stick
- Low confidence with new technology
- Wary of high prices and batteries that fail in the cold

Fictional persona built from our research; to be refined after interviews.

The same task, in two different worlds

	Before: current routine	After: Smart Mailbox + app
Effort	Daily 80 m walk; most of the time the box is empty	Glance at the phone; walk out only when mail is there
Timeliness	Checks postponed for days; bills and notices sit unseen	Alert the moment mail drops in; collected on time
Safety & independence	Risky walks "just in case", or asks son / neighbour	Decides herself when to go, in daylight and safe conditions

Proposed future-state journey: checking mail with the app

Step	What Aino does	What the product does	Intended improvement
1. Receive alert	Gets a notification on her phone, at home or away.	Detects that mail is inside the box; app sends *"New mail has arrived."*	No guessing, no walks on suspicion.
2. Check status	Opens the app, decides when to walk out.	Shows arrival time and system status (battery).	No empty walks in bad weather.
3. Retrieve mail	Walks out on her own schedule and collects letters.	Keeps showing "Mail arrived" as long as mail is still inside.	Every trip counts; the status never misleads.
4. Auto-reset	Back indoors, sees the app now shows "Empty".	Switches to "Empty" as soon as the mail is taken out, ready for the next delivery.	Zero manual hassle.

Why this journey?	What we will test in the prototype
Checking the mailbox is Aino's most frequent and most uncertain mail task, and where the effort and risk happen. The journey shows the full loop from need to outcome.	**1. Trust:** Is a simple "Mail arrived" alert enough for users to trust it, or do unwanted flyers cause frustrating false trips? **2. Detail:** Do users want to know the type or amount of mail? **3. Ease of use:** Is the app simple enough that older, non-technical users rely on it every day?

Other relevant people

Stakeholder	Role and interest
Family members	Worry about safety; may install, set up or pay for the product.
Home care workers	Visit vulnerable residents; may help with the device.
Neighbours and friends	Today's informal backup; the product must not depend on them.
Posti delivery staff	Deliver as usual; no extra work on their routes.
Senders (billers, authorities)	Gain when important letters are read on time.

3. Business Model

Model in brief: Phase 1 (launch): sell a one-time hardware kit plus a small recurring fee directly and through local retail to older residents and their adult children. **Phase 2 (scale):** add B2B partnerships and subsidies with municipalities, eldercare services and home insurers interested in fall prevention. Pricing and partner agreements follow in later phases.

Business Model Canvas	Proposed choice
Customer segments	**End users:** older adults (65+), people with reduced mobility, rural residents far from their roadside mailbox. **Phase 1 payers:** the residents themselves and their adult children. **Phase 2 sponsors:** municipalities, eldercare services and home insurers (fall prevention).
Value proposition	Removes unnecessary, risky walks to an empty mailbox through instant indoor awareness of delivery, regardless of weather, distance or mobility, without advanced smartphone skills or extra postal work.
Channels	**Phase 1:** direct online sales; local Finnish hardware and retail stores; social media aimed at older residents and their adult children; app stores for the companion app. **Phase 2:** B2B partnerships with municipalities, eldercare services and home insurers.
Customer relationships	Simple self-installation or assisted setup (family, home care, maintenance staff); automatic battery and status alerts in the app; phone support.
Revenue streams / funding	**Phase 1:** one-time hardware kit purchase + small recurring fee covering the mobile connection and support. **Phase 2:** B2B bulk sales and subsidies from municipalities, eldercare services and home insurers.
Key activities	Product design for reliable all-weather outdoor use (with manufacturing partners); app development and updates; customer support; retail sales and marketing (Phase 1); building institutional partnerships (Phase 2).
Key resources	Mailbox module and companion app; built-in mobile connection; customer feedback and usage insights; support team; retail relationships (Phase 1) and institutional relationships (Phase 2).
Key partners	Electronics manufacturers; mobile network providers; local retailers; local maintenance services. Phase 2: municipalities, eldercare services and home insurers.
Cost structure	Manufacturing and assembly; product certifications; mobile connection fees; app maintenance and updates; marketing, retail margins and customer support.

How the model fits together

Key connection	What it means
Users vs. payers	Aino uses it. In Phase 1 she or her son buys it and sets it up. In Phase 2, municipalities, eldercare services or insurers may subsidise it because fewer risky walks means fewer falls and longer independent living.
How value reaches users	Phase 1: direct and local retail sales, promoted through social media to residents and adult children. Phase 2: eldercare services and insurers bring it to vulnerable residents, with assisted setup. Daily value arrives via app alerts.
Critical dependencies	Reliability is the promise: correct alerts in all weather, long battery life and a working connection at the roadside (manufacturing and mobile network partners).
Why phase it?	Phase 1 proves demand and reliability with paying households (kit covers hardware; small fee covers connection and support). Phase 2 uses that evidence to win partners whose subsidies lower the price for at-risk users like Aino.

Conclusion and Next Steps

> We have built one focused concept that links the problem found in our research to the daily reality of our users and to a realistic business model.
>
> The Smart Mailbox replaces a habit-driven, often empty and sometimes risky walk with a simple, reliable signal: mail has arrived, or it has not.
>
> The offer in Section 1, Aino's journey in Section 2 and the business model in Section 3 describe the same product, the same users and the same core value.

Next step	Purpose
Landing page	Communicate the promise to residents and their families.
Functional prototype	Demonstrate the critical journey: alert → status check → retrieval → auto-reset.
Usability and trust testing with older users	Test whether the simple "Mail arrived" / "Empty" alert is easy to understand, builds daily trust, and replaces empty trips (including any false trips caused by flyers).

AI use: AI Assistant (Claude by Anthropic) helped merge, edit and format the team's section drafts. All content was reviewed by the team.